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Assignment -5

1) Compare TCP and UDP Protocols.

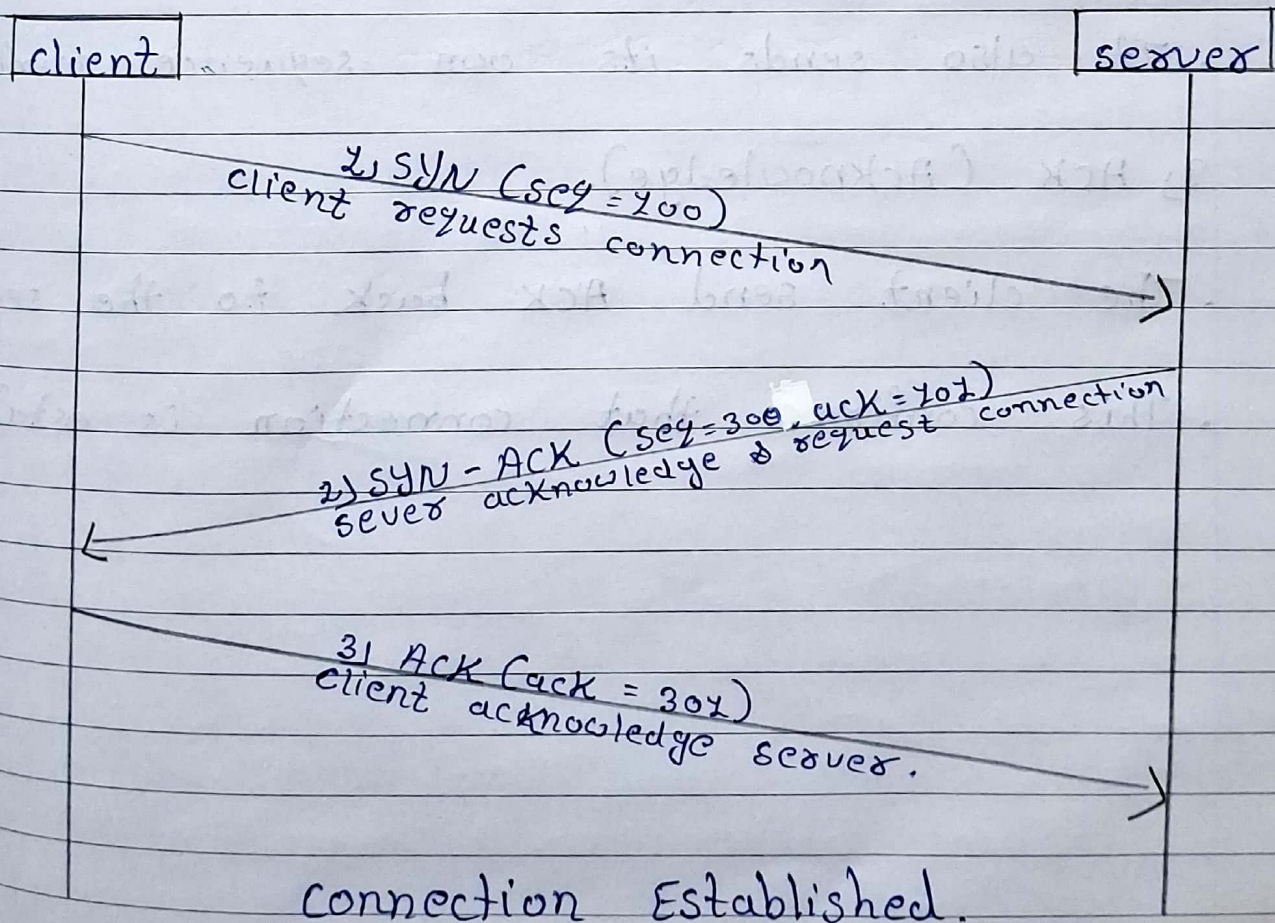
→ TCP	UDP
1) TCP stand for (Transmission control Protocol)	UDP stand for (User Datagram Protocol)
2) TCP support error-checking mechanisms.	UDP has only the basic error-checking mechanism
3) An Acknowledge segment is present	No acknowledge segment
4) TCP is slower than UDP	UDP is faster
5) Retransmission of lost Package is possible in TCP	There is no retransmission of lost packets in the UDP
6) TCP has a (20-60) bytes variable length header.	The header length is Fixed of 8 bytes.

2) Explain the TCP 3-way handshake process with a neat diagram.

→ The TCP 3-way handshake is a process used in the Transmission Control Protocol (TCP) to establish a reliable connection between a client and a server before data transfer begins.

It ensures that both sides are ready to communicate and synchronize sequence numbers.

Diagram



1) syn (synchronize)

- The client send a syn Packet to the server
- It indicates the client wants to start a ~~comm~~ connection.
- It also send an initial sequence number.

2) syn-ACK (synchronize - Acknowledge)

- The server responds with syn + ACK
- It acknowledge the client's request
- It also sends its own sequence number.

3) ACK (Acknowledge).

- The client send ACK back to the server.
- This confirms that connection is established.

3) Describe congestion ^{control} techniques used in TCP.

→ Congestion control in TCP is used to prevent ~~too~~ too much data from being sent into the network, which could overload routers and cause packet loss, delay, or network collapse.

Congestion control Technique.

open loop congestion control

closed loop congestion control

1) Open loop congestion control

open loop congestion control tries to prevent congestion before it happens. It does not rely on feedback from the network after congestion occurs.

Techniques used:

i) Retransmission Policy → Avoid unnecessary retransmissions.

ii) Window Policy → control number of packets sent

- iii) Acknowledgment Policy $\rightarrow$ Reduce ACK traffic.
- iv) Admission Policy $\rightarrow$ Allow only limited traffic into network.
- v) Discard Policy $\rightarrow$ drop less important packets when needed.

Idea:- "Prevent congestion in advance".

2) close loop congestion

close loop congestion control works by detecting congestion and then taking corrective action using feedback from the network.

Techniques used:

- i) Backpressure $\rightarrow$ congested node tells previous nodes to slow down
- ii) choke packets $\rightarrow$ Router sends special packet to reduce sending rate.
- iii) Implicit signaling $\rightarrow$ sender detects congestion via packet loss or delay
- iv) Explicit signaling (ECN) $\rightarrow$ network explicitly informs sender about congestion.

Idea:- "control congestion after it occurs".

4) Explain the Domain Name system (DNS) and its working process.

→ To Domain Name system (DNS) is a naming system used in computer networks to convert human-readable domain name into IP addresses.

computers understand IP address (like 74.250.190.74), but human prefer names (like www.google.com). DNS acts like a phonebook of the Internet.

DNS Working Process

When you enter a website name in your browser, DNS helps to find its correct IP address so the website can load.

1) Enter Website Name

you type a domain name like www.google.com in the browser.

computers cannot understand names, so they need an IP address.

2) Browser checks Cache

The browser first checks its local cache (stored memory) to see if it already knows the IP address.

If found, it directly opens the website (Fast Process).

3] DNS Resolver & request

If not found, the request is sent to a DNS Resolver (usually ISP server).

It acts as a middleman between browser and DNS system.

4] Root DNS server

The resolver contacts the Root DNS server. Root server does not know the exact IP, but it guides to the correct next level.

5] Authoritative DNS server

This server contains the actual information of the domain.

If finds and returns the correct IP address of website.

7] IP Address Returned

The resolver sends the IP address back to the browser.

8) Website opens

Now the browser uses the IP address to connect to the web server and load the website.

DNS work step-by-step to convert a domain name into an IP address so that browser can find and open the website.

5) Describe the File Transfer Protocol (FTP) and explain its working.

→ File Transfer Protocol (FTP) is an application layer protocol used to transfer files between a client and a server over a network.

It allows users to upload, download, delete, rename, and manage files remotely on a server.

Working of FTP

FTP work using two separate communication channels:

- control channel:- used to send commands (like login, list files, download request)
- data channel:- used to transfer actual file data.

4) Connection Establishment

The FTP client first connects to the FTP server using the server's IP address or domain name.

This connection is made through the control channel.

2) Login Process

The user must authenticate by providing a username and password.

Some servers allow anonymous FTP, where users can access files without a login.

3) Command communication

After login, the client sends commands to the server through the control channel, such as:

- List files
- Change directory
- Download file
- Upload file.

4) Data Transfer

When a file transfer is requested;

- A separate data connection is created.
- The file is transferred through this channel.
- After completion, the data connection is closed.

5) closing connection

once all operation are done, the client sends a quit command, and both channel control and data connection are closed,

6) Explain simple Mail Transfer Protocol (SMTP) and its role in email communication.

-) simple Mail Transfer Protocol (SMTP) is a protocol used to send emails over the internet. From a sender to a mail server and between mail servers. It only handles sending emails, not receiving them.

Role of SMTP in Email communication

SMTP plays a key role in ensuring that emails are delivered from the sender to the recipient correctly and reliably.

1) sending Email from client

When you send an email using application like Gmail or Outlook:

- The email client uses SMTP to send the message to the mail server.

2) Mail server Processing

The sender's mail server:

- checks the recipient's address
- Determines the destination server using domain information.

- Prepares the email for transfer.

3) Transfer Between servers

SMTP is used again to:

- Transfer the email from the sender's server to recipient's mail server.
- This may involve multiple intermediate mail server.

4) Delivery to Recipient server

The email finally reaches the recipient's mail server, where it is stored until the user accesses it.

Note :- SMTP does not retrieve emails. Protocols like IMAP or POP3 are used for receiving emails.

7) Define HTTP and HTTPS. List the key differences between them.

	HTTP	HTTPS
Full form	Hypertext Transfer Protocol	Hypertext Transfer Protocol secure.
Security	Not secure	Secure (uses encryption)
Data protection	Data is sent in Plain text	Data is encrypted
Port number	80	443
Certificate	Not required certificate	Requires SSL/TLS certificate.
Speed	Faster	slightly slower due to encryption.
Usage	Basic Websites	Banking, login Pages, secure sites.
URL	URL begins With "http://"	URL start with "https://"

8) Explain the architecture of an email system with components like user agent, mail agent, server, and protocols.

→ Email system Architecture

An email system works in a client-server model and mainly consists of User Agent, Mail server, and protocols.

1) User Agent (Mail client)

It is the software used by the user to access emails.

- Used to compose, send, and read emails
- Examples: Gmail, Outlook, Thunderbird,
- Act as the interface between user and email system.

2) Mail server

It handles storage and transfer of emails between sender and receiver.

- SMTP server (Outgoing server): sends emails from sender to recipient's mail server.
- POP3 / IMAP server (Incoming server): ~~:- sends~~ stores and manages received emails.

- POP3 downloads emails to device.
- IMAP keeps emails on server for multi-device access.

3) Email Protocols

Protocols define rules for email communication.

- SMTP : used to sending email
- POP3 : used for downloading emails to local system.
- IMAP : Used for accessing emails directly from server.